A VQE Study of the Lipkin Interaction

FYS5419 – Quantum computing and quantum machine learning

PROJECT 1

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Abstract

Financial markets exhibit pronounced non-stationarities driven by structural breaks, macroeconomic shocks, and evolving investor behavior. This project investigates the use of machine learning methods for identifying and modeling regime changes in financial time series. We consider supervised and unsupervised approaches to regime detection, including clustering-based methods and probabilistic state models, and evaluate their ability to capture shifts in volatility, return dynamics, and cross-asset dependencies. Model performance is assessed using historical market data under realistic out-of-sample settings. The results highlight both the potential and limitations of machine-learning-based regime modeling in quantitative finance, with implications for risk management, portfolio allocation, and adaptive trading strategies.

1 Introduction

2 Methods

2.1 Part A methods

One-Qubit bases

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix} \quad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix} \tag{1}$$

Pauli Matrices

$$\sigma_x = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix} \quad \sigma_y = \begin{pmatrix}0 & -i\\i & 0\end{pmatrix} \quad \sigma_z = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix} \tag{2}$$

Hadamard and Phase Gates

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix} \quad S = \begin{pmatrix}1 & 0\\0 & -i\end{pmatrix} \tag{3}$$

Kronecker products for basis states

$$|00\rangle = |0\rangle \otimes |0\rangle \quad |01\rangle = |0\rangle \otimes |1\rangle \tag{4}$$

$$|10\rangle = |1\rangle \otimes |0\rangle \quad |11\rangle = |1\rangle \otimes |1\rangle \tag{5}$$

Bell states

$$\left|\Phi^+\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad \left|\Phi^-\right\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \tag{6}$$

$$\left|\Psi^+\right\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \quad \left|\Psi^-\right\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} \tag{7}$$

$$\text{CNOT} = \begin{pmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 0 & 1\\0 & 0 & 1 & 0\end{pmatrix} \tag{8}$$

2.2 Part B methods

$$H = H_0 + H_1 = \begin{bmatrix}H_{11} & H_{12}\\H_{21} & H_{22}\end{bmatrix} \tag{9}$$

$$H_0 = \begin{bmatrix}E_1 & 0\\0 & E_2\end{bmatrix} \quad H_1 = \begin{bmatrix}V_{11} & V_{12}\\V_{21} & V_{22}\end{bmatrix} \tag{10}$$

Where V_{ij} represents various interaction matrix elements. We can view H_0 as the non-interacting solution

$$H_0 |0\rangle = E_1 |0\rangle \quad H_0 |1\rangle = E_2 |1\rangle \tag{11}$$

We rewrite H with H_0 and H_1 via Pauli matrices

$$H_0 = \mathcal{E} I + \Omega \sigma_Z \quad H_I = c\mathbf{I} + \omega_z \sigma_z + \omega_x \sigma_x \tag{12}$$

$$\mathcal{E} = \frac{E_1 + E_2}{2} \quad \Omega = \frac{E_1 - E_2}{2} \quad \mathbf{I} = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} \tag{13}$$

$$c = \frac{V_{11} + V_{22}}{2} \quad \omega_z = \frac{V_{11} - V_{22}}{2} \quad \omega_x = V_{12} = V_{21} \tag{14}$$

We let our Hamiltonian depend linearly on a strength parameter λ

$$H = H_0 + \lambda H_1 \tag{15}$$

with $\lambda \in [0, 1]$ and limits $\lambda = 0$, $\lambda = 1$.

2.3 Part C methods

2.4 Part D methods

$$|0\rangle_{A,B} = [1 \quad 0]^T \quad |1\rangle_{A,B} = [0 \quad 1]^T \tag{16}$$

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B = [1 \quad 0 \quad 0 \quad 0]^T \tag{17}$$

$$|01\rangle = |0\rangle_A \otimes |1\rangle_B = [0 \quad 1 \quad 0 \quad 0]^T \tag{18}$$
$$|10\rangle = |1\rangle_A \otimes |0\rangle_B = [0 \quad 0 \quad 1 \quad 0]^T \tag{19}$$
$$|11\rangle = |1\rangle_A \otimes |1\rangle_B = [0 \quad 0 \quad 0 \quad 1]^T \tag{20}$$

$$H_0 |00\rangle = \epsilon_{00} |00\rangle \tag{21}$$
$$H_0 |10\rangle = \epsilon_{10} |10\rangle \tag{22}$$
$$H_0 |01\rangle = \epsilon_{01} |01\rangle \tag{23}$$
$$H_0 |11\rangle = \epsilon_{11} |11\rangle \tag{24}$$

$$H_I = H_x \sigma_x \otimes \sigma_x + H_z \sigma_z \otimes \sigma_z \tag{25}$$

$$\mathbf{H} = \begin{bmatrix} \epsilon_{00} + H_z & 0 & 0 & H_x \\ 0 & \epsilon_{10} - H_z & H_x & 0 \\ 0 & H_x & \epsilon_{01} - H_z & 0 \\ H_x & 0 & 0 & \epsilon_{11} + H_z \end{bmatrix} \tag{26}$$

$$\rho_0 = (\alpha_{00} |00\rangle \langle 00| + \alpha_{10} |10\rangle \langle 10| + \alpha_{01} |01\rangle \langle 01| + \alpha_{11} |11\rangle \langle 11|) \tag{27}$$
$$\rho_A = \mathrm{Tr}_B(\rho_0) = \langle 0| \rho_0 |0\rangle_B + \langle 1| \rho_0 |1\rangle_B \tag{28}$$
$$\rho_B = \mathrm{Tr}_A(\rho_0) = \langle 0| \rho_0 |0\rangle_A + \langle 1| \rho_0 |1\rangle_A \tag{29}$$
$$S(A, B) = -\mathrm{Tr}(\rho_{A,B} \log_2(\rho_{A,B})) \tag{30}$$

2.5 Part E methods

2.6 Part F methods

$H = H_0 + H_1 + H_2$

$$H_0 = \frac{1}{2}\varepsilon \sum_{p\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma} \tag{31}$$

$$H_1 = \frac{1}{2} V \sum_{p,p',\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma} \tag{32}$$

$$H_2 = \frac{1}{2} W \sum_{p,p',\sigma} a^\dagger_{p\sigma} a^\dagger_{p'\sigma} a_{p'-\sigma} a_{p-\sigma} \tag{33}$$

$$H_0 = \varepsilon J_z \tag{34}$$

$$H_1 = \frac{1}{2} V (J_+^2 + J_-^2) \tag{35}$$

$$H_2 = \frac{1}{2} W (-N + J_+ J_- + J_- J_+) \tag{36}$$

$$H_{J=1} = \begin{pmatrix} -\epsilon & 0 & -V \\ 0 & 0 & 0 \\ -V & 0 & \epsilon \end{pmatrix} \tag{37}$$

For $J = 2$ we have a 5×5 matrix given by

$$H_{J=2} = \begin{pmatrix} -2\epsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\epsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \epsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\varepsilon \end{pmatrix} \tag{38}$$

2.7 Part G methods

3 Results

3.1 Part A results

3.2 Part B results

Listing 1. VQE solver loop

```
# Solve for lambda in [0, 1]
lambdas = np.linspace(0, 1, 101)
eigvals = []
eigvecs = []

for l in lambdas:
    H = H0 + l * HI
    w, v = np.linalg.eigh(H)
    eigvals.append(w)
    eigvecs.append(v)

eigvals = np.array(eigvals)
eigvecs = np.array(eigvecs)
```

3.3 Part C results

$$E_1 = 0.0 \quad E_2 = 4.0 \tag{39}$$
$$V_{11} = 3.0 \quad V_{12} = 0.2 \tag{40}$$
$$V_{21} = 0.2 \quad V_{22} = -3.0 \tag{41}$$
$$H_0 = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix} \tag{42}$$
$$H_I = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix} = \begin{pmatrix} 3.0 & 0.2 \\ 0.2 & -3.0 \end{pmatrix} \tag{43}$$

$$\mathcal{E} = 2.0 \quad \Omega = -2.0 \quad \omega_z = 3.0 \quad \omega_x = 0.2 \tag{44}$$

$$H(\lambda) = H_0 + \lambda H_1 \tag{45}$$

$$H(\lambda)\mathbf{v}_i = E_i(\lambda)\mathbf{v}_i \tag{46}$$

$$H(\lambda) = \varepsilon \mathbf{I} + (\lambda\omega_z + \Omega)\sigma_z + (\lambda\omega_x)\sigma_x \tag{47}$$

$$|\psi(\theta)\rangle = R_y(\theta) |0\rangle \tag{48}$$

$$|\psi(\theta)\rangle = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) \end{pmatrix} \tag{49}$$

$$\langle \sigma_z \rangle = \cos(\theta) \quad \langle \sigma_x \rangle = \sin(\theta) \tag{50}$$

$$E(\theta, \lambda) = \varepsilon + (\lambda\omega_z + \Omega)\cos(\theta) + (\lambda\omega_x)\sin(\theta) \tag{51}$$

$$E_{\mathrm{VQE}}(\lambda) = \min_{\theta} E(\theta, \lambda) \tag{52}$$

$$E_{\mathrm{exact}}(\lambda) \tag{53}$$

$$\lambda_i \in [0, 1] \quad i = 1, \ldots, 51 \tag{54}$$

3.4 Part D results

$$H_x = 2.0 \quad H_z = 3.0 \tag{55}$$

$$|\epsilon\rangle_{00} = 0.0 \quad |\epsilon\rangle_{10} = 2.5 \tag{56}$$

$$|\epsilon\rangle_{01} = 6.5 \quad |\epsilon\rangle_{11} = 7.0 \tag{57}$$

3.5 Part E results

3.6 Part F results

3.7 Part G results

4 Conclusion

4.1 Sample-text

Quantum computing builds on fundamental principles of quantum mechanics such as superposition, interference, and entanglement, concepts that were already central to early discussions about the completeness of quantum theory (Einstein et al. 1935). Modern treatments of quantum computation are typically grounded in the formal framework presented by Nielsen and Chuang (2000), while more accessible introductions can be found in texts such as Rieffel and Polak (2011) and in pedagogical overviews aimed at non-physicists (Rieffel et al. 2000). From a mathematical perspective, the structure of quantum algorithms and Hilbert space methods is explored in detail by Scherer (2018), with complementary textbook treatments provided in alternative presentations of the quantum information framework (Michaelsen and Isaac 2019).

Within this broader landscape, the emerging field of quantum machine learning (QML) seeks to combine ideas from classical statistical learning with quantum algorithms. Classical foundations for statistical learning theory and modern machine learning methods are well established in works such as Hastie et al. (2009), which provide the theoretical backbone for many learning algorithms. Building on these principles, several authors have developed dedicated treatments of QML, including comprehensive overviews (Conti 2020; Schuld 2021) and more practical programming-oriented introductions (Pattanayak et al. 2021). Current algorithmic research often focuses on hybrid quantum–classical methods designed for near-term quantum devices. For instance, the Variational Quantum Eigensolver (VQE) has emerged as one of the most widely studied approaches for optimization and quantum simulation problems, with extensive methodological surveys available in the literature (Tilly et al. 2022). Closely related ideas appear in the Quantum Approximate Optimization Algorithm (QAOA), whose theoretical performance and practical implementation on near-term hardware have recently been analyzed in detail (Zhou et al. 2025), with introductory expositions available in recent tutorial-style papers (Giovagnoli 2026).

In addition to traditional academic literature, open educational resources and code repositories have become an important part of the quantum computing ecosystem. For example, the publicly available course repository by Hjorth-Jensen, Melkild, et al. (2026) provides practical implementations and teaching material covering both quantum algorithms and quantum machine learning methods.

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5 Appendix

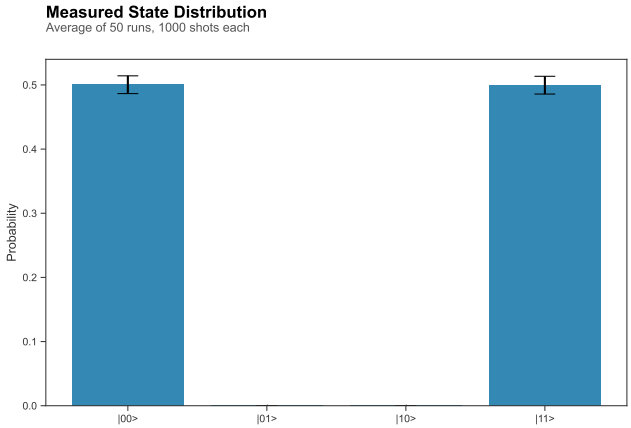

Figure 1. Measurement distribution (Part A).

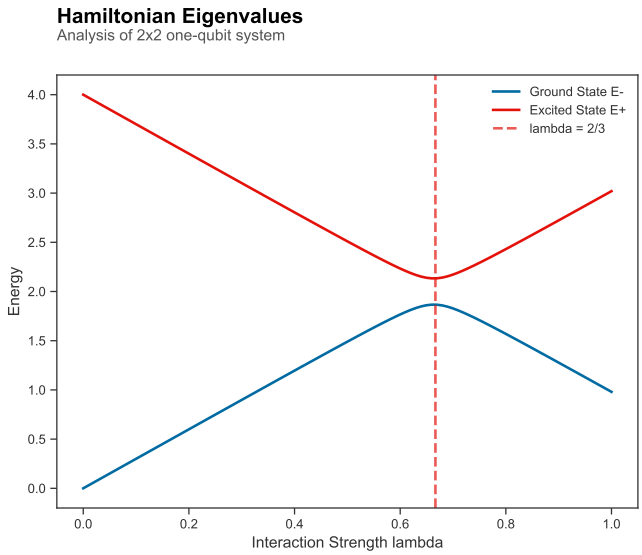

Figure 2. Eigenvalue spectrum of the Hamiltonian (Part B).

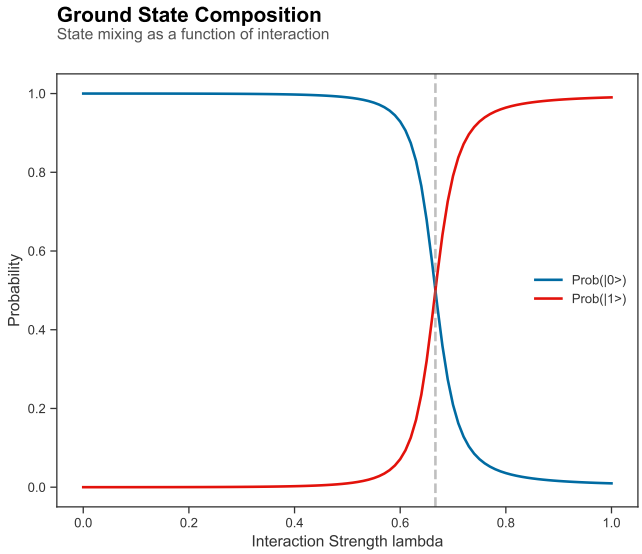

Figure 3. Eigenvector mixing across the spectrum (Part B).

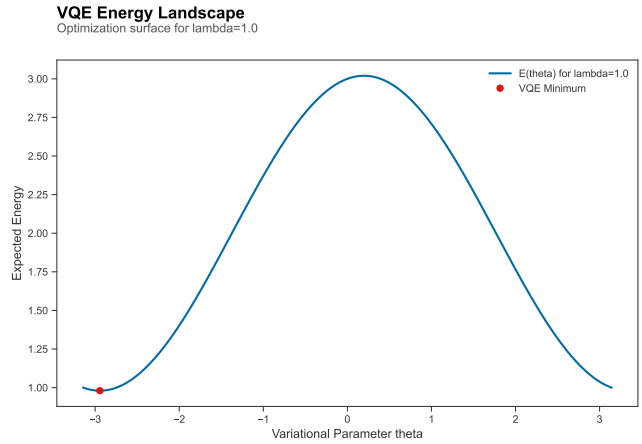

Figure 4. Energy landscape over the variational parameter space (Part C).

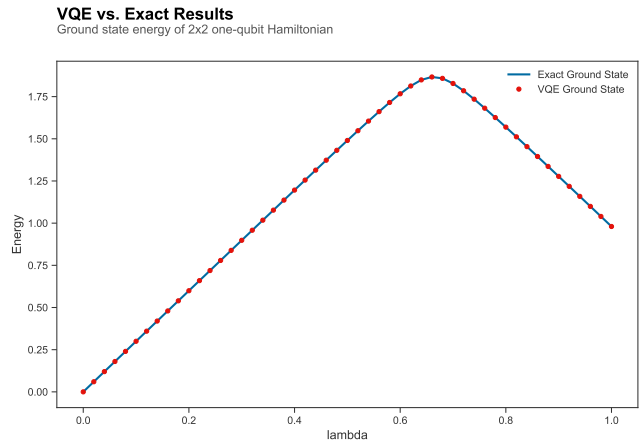

Figure 5. VQE energy convergence compared to the exact ground-state energy (Part C).

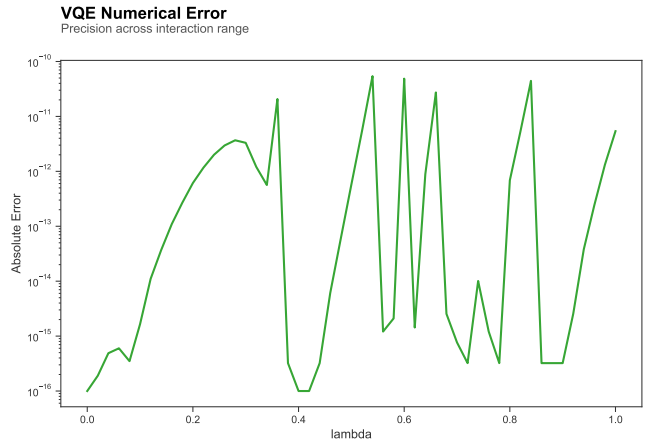

Figure 6. Absolute VQE error as a function of optimization iterations (Part C).

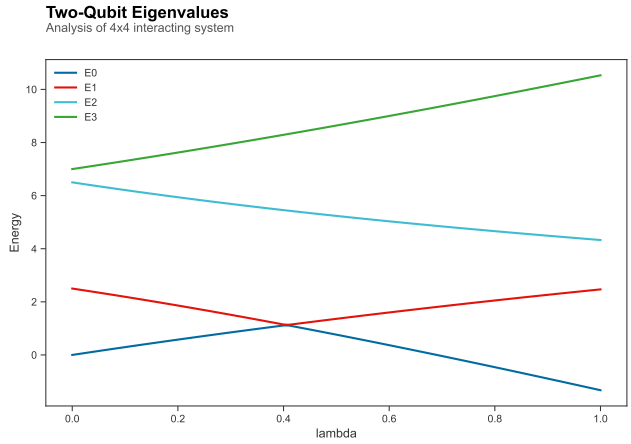

Figure 7. Eigenvalue spectrum of the extended system (Part D).

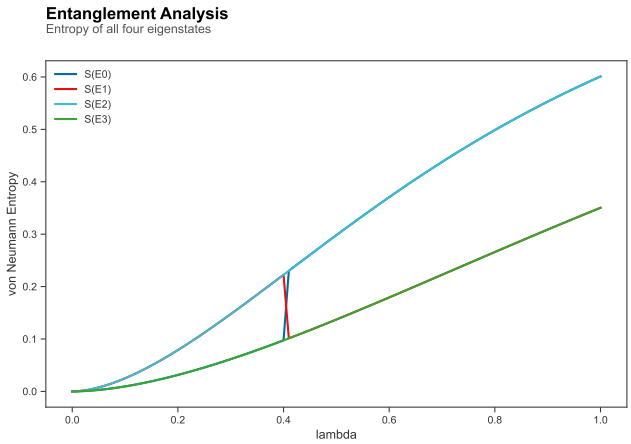

Figure 8. Bipartite entanglement entropy across system partitions (Part D).

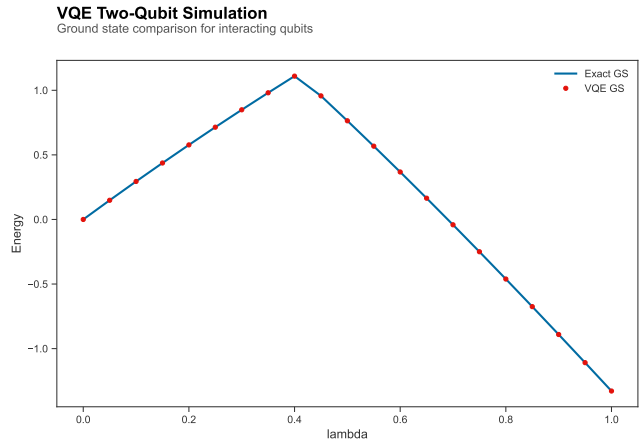

Figure 9. VQE energy convergence for the modified Hamiltonian (Part E).

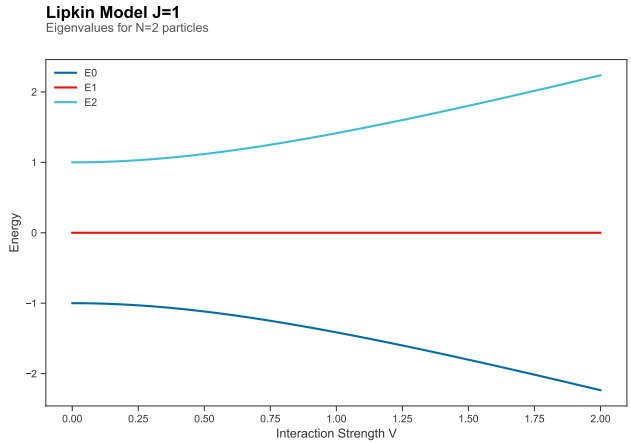

Figure 10. Ground-state energy as a function of coupling parameter J_1 (Part F).

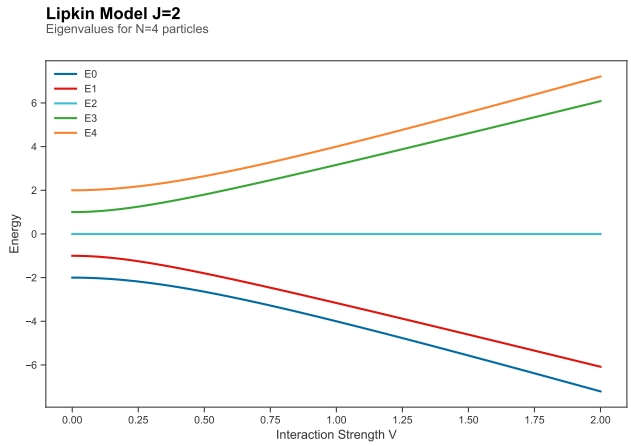

Figure 11. Ground-state energy as a function of coupling parameter J_2 (Part F).

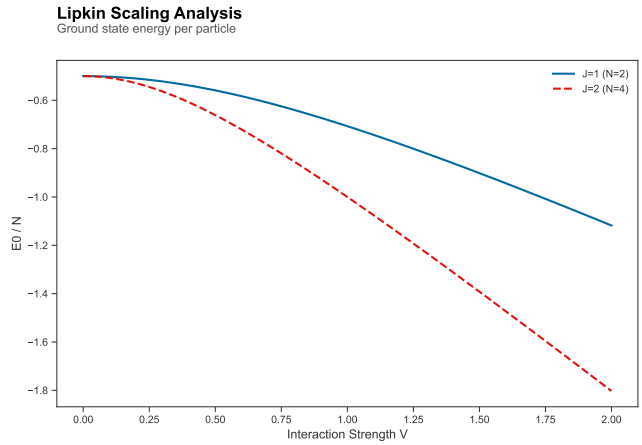

Figure 12. Scaling behavior of the ground-state energy with system size (Part F).

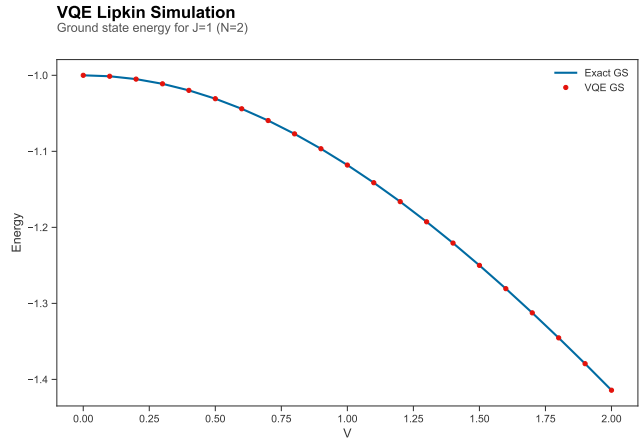

Figure 13. VQE optimization landscape under variation of J_1 (Part G).